

Evidence-aligned immunometabolic analysis in human intracerebral hemorrhage

Verified data types, corrected longitudinal unit of analysis, sensitivity tests, and interpretation boundary

a Verified datasets

GSE163256

Longitudinal sorted-cell transcriptomics
CD14+ monocytes/macrophages
and neutrophils
21 ICH patients + 5 healthy controls



Locked monocyte analysis subset

136 unique ICH observations
20 patients; days 1–6
technical duplicates collapsed

GSE166638

scRNA-seq from one longitudinally
sampled patient plus follow-up/control blood
exploratory cellular triangulation only

b Corrected primary models

Patient-level mixed models

Inflammation: $\beta = 0.319$ (95% CI 0.131 to 0.507)
Hypoxia: $\beta = 0.831$ (95% CI 0.690 to 0.972)

Model: $\text{outcome} \sim \text{Index} + \text{compartment} + \text{day} + (1 \mid \text{patient})$

Overlap-pruned sensitivity

Inflammation: $\beta = 0.372$ (95% CI 0.144 to 0.601)
Hypoxia: $\beta = 0.647$ (95% CI 0.450 to 0.845)

Overlap-pruned gene sets removed all cross-signature overlaps

c Interpretation boundary

Supported

Expression-derived metabolic scores
co-vary with hypoxia and inflammation
in longitudinal sorted
monocyte profiles



Not established

Metabolic flux or causal niche effects
population-level scRNA-seq
replication or therapeutic actionability

CellChat and virtual-knockout results from GSE166638 are retained only as hypothesis-generating candidate nomination.

Index denotes mean Hallmark glycolysis expression minus mean Hallmark oxidative-phosphorylation expression.